Practical – 10

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Aim:

To understand and implement version control using Git and GitHub.

Theory:

Version Control is a system that helps track changes in files over time. It allows multiple users to work on a project and keeps a history of changes.

Git is a distributed version control system used to track changes locally.
GitHub is an online platform used to store and manage Git repositories remotely.

Using Git and GitHub, we can:

- Track file changes
- Collaborate with others
- Maintain version history
- Restore previous versions

⚙️ **Step-by-Step Procedure**

Step 1: Installation Verification

Command used:

git --version

Git installation was verified successfully.

Step 2: Configure Username & Email

Commands:

git config --global user.name "Sarang06"
git config --global user.email "sarangchvan06@gmail.com"

Step 3: Create Local Repository

Commands:

mkdir myproject
cd myproject
git init

A new Git repository was initialized.

Step 4: Create File

Command:

echo Hello Git > file1.txt

A new file was created in the repository.

Step 5: Add File to Staging Area

Command:

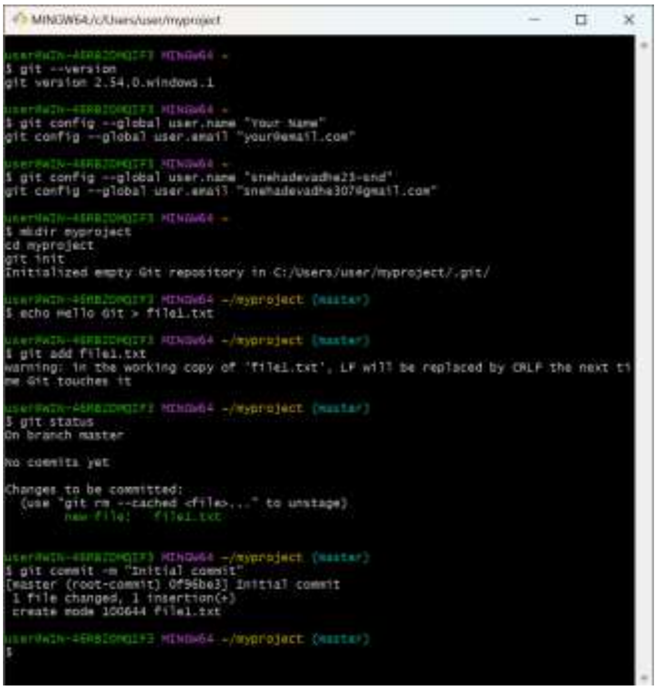
git add file1.txt

Step 6: Commit Changes

Command:

git commit -m "Initial commit"

Changes were successfully committed.



Step 7: Create GitHub Repository

A new repository named **PDS_PRACTICALS** was created on
GitHub

Step 8: Link Local Repository to GitHub

Command:

git remote add origin https://github.com/Sarango6/PDS_PRACTICALS.git

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Step 9: Push Files to GitHub

Commands:

git branch -M main
git push -u origin main

The files were successfully uploaded to GitHub.

```
user@WIN-46RB2DMQIF3 MINGW64 ~
$ git config --global user.name "snehadevadhe23-snd"
git config --global user.email "snehadevadhe307@gmail.com"

user@WIN-46RB2DMQIF3 MINGW64 ~
$ mkdir myproject
cd myproject
git init
Initialized empty Git repository in C:/Users/user/myproject/.git/

user@WIN-46RB2DMQIF3 MINGW64 ~/myproject (master)
$ echo Hello Git > file1.txt

user@WIN-46RB2DMQIF3 MINGW64 ~/myproject (master)
$ git add file1.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next ti
me Git touches it

user@WIN-46RB2DMQIF3 MINGW64 ~/myproject (master)
$ git status
On branch master
Merge branch 'main' of https://github.com/snehadevadhe23-snd/PDS_PRARCTICAL
```

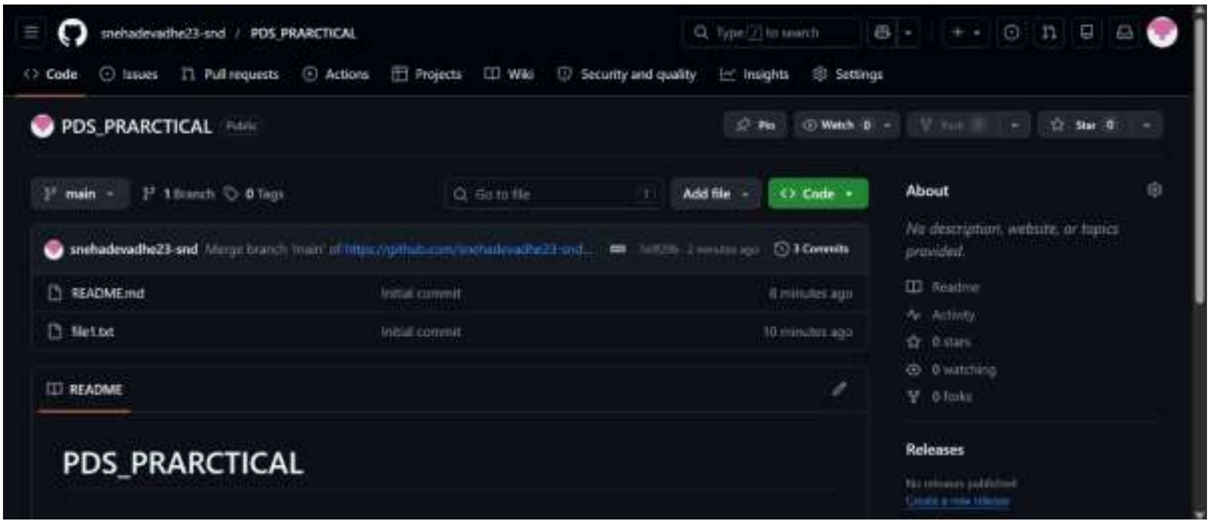
Step 10: Verify Files on GitHub

The file **file1.txt** is visible in the GitHub repository.

Step 11: Display Commit Logs

Command:

git log



Questions & Answers

1. What is the difference between Git and GitHub?

- **Git:** A version control system used to track changes in files locally.
- **GitHub:** A cloud-based platform used to store and manage Git repositories online.

2. Difference between git add and git commit

- **git add:** Adds changes to the staging area.
- **git commit:** Saves changes permanently in the repository.

3. What is the staging area in Git?

The staging area is a temporary space where changes are prepared before committing them to the repository.

4. What happens if you modify a file after committing it?

Git detects it as a modified file. You must again use:

git add
git commit

to save the new changes.

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5. Difference between git push and git pull

- **git push:** Uploads local changes to GitHub repository.
- **git pull:** Downloads latest changes from GitHub to local system.

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6. What is Version Control? Differentiate between Centralized and Distributed VCS

Version Control is a system that tracks changes in files over time.

Centralized VCS:

- Single central server
- Example: SVN
- Users depend on server

Distributed VCS:

- Each user has full copy of repository
- Example: Git
- Works even without internet

Final Conclusion

This practical successfully demonstrated the use of Git and GitHub for version control. All steps including repository creation, staging, committing, and pushing to GitHub were performed correctly. The experiment helped in understanding how version control systems manage project history and enable collaboration.